

INVENTORY COORDINATION SYSTEM

Domain 1: Agentic Workflow & Autonomous Coordination



THE PROBLEM: THE COORDINATION TAX

*The hidden operational cost paid in time, labor, and errors
whenever a human must manually 'bridge' the gap between disconnected tools.*

Fragmented Data

Data is scattered
across silos
(Inbox vs Spreadsheet)

Manual Process

Requires constant
human interpretation &
data entry

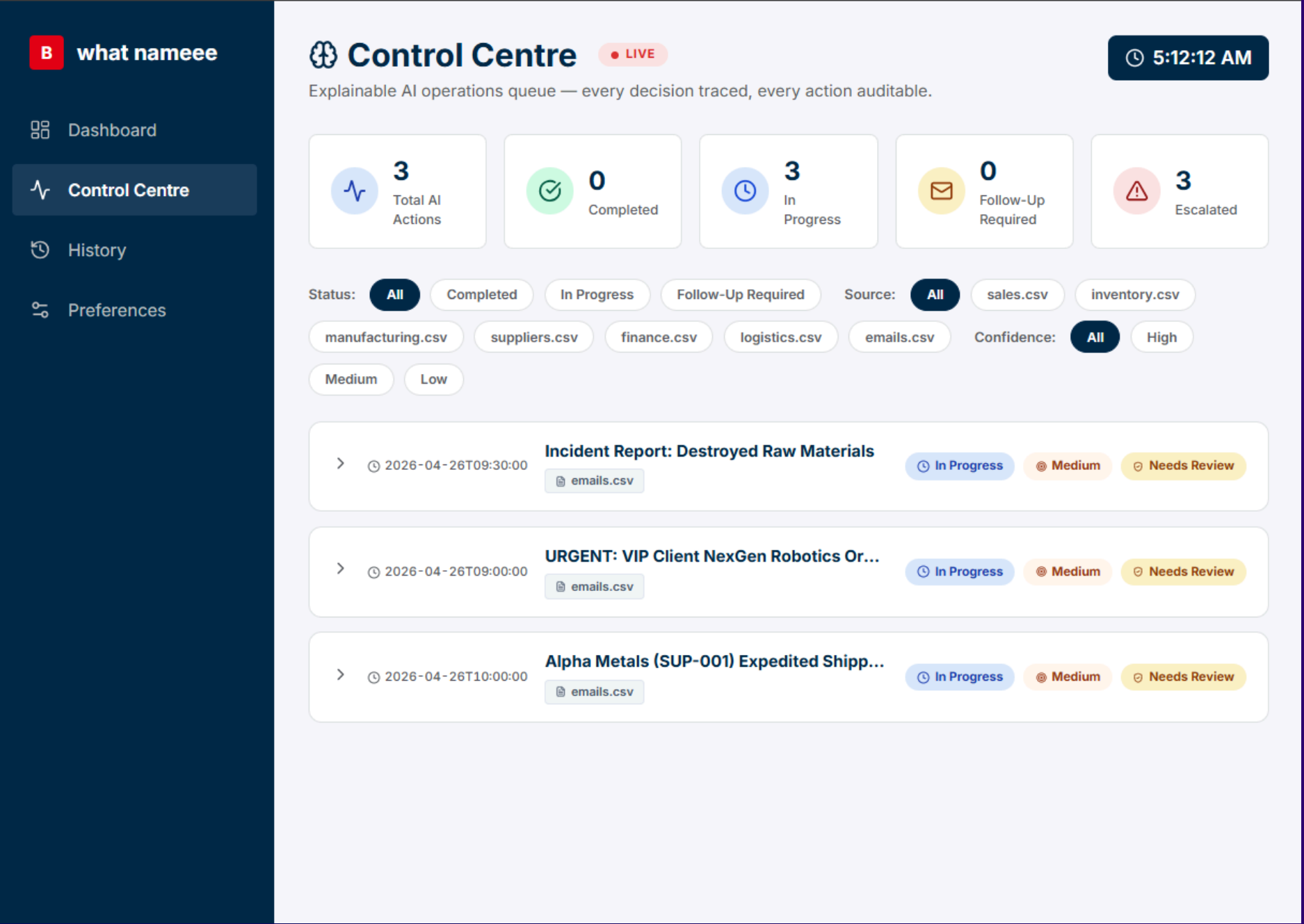
Unstructured Inputs

"Messy Input" (emails)
cannot be read by
traditional software

The Result

**A 30% Coordination Tax on operational efficiency &
hard ceiling on company growth**

This is the 'Hard Ceiling' that stops a Malaysian SME from becoming a global player.



UI mockup of the platform

Moving from
**HUMAN-LED
COORDINATION**
to
**AUTONOMOUS
AI EXECUTION**

THE UNIFIED TRUTH

We meet businesses where they are, using the tools they already have.

INPUT LAYER

**Unstructured Emails &
CSV Spreadsheets**

REASONING LAYER

**Gemini GLM shows:
contextual
understanding &
performs multi-step
logic**

ACTION LAYER

**Re-allocate stocks &
Automated Purchase
Orders & Production
Schedules**

THE INPUT

Current Date: **1st April**

Current Stock: **100 units SKU-A | 0 units Raw Material**

The Conflict:

Commitment A

Order #1 : 200 units
Delivery: **April 20**

*Requires additional 100 units
+
50 units Raw materials*

Urgent Request B

Order #2: 10 units
Delivery: **TODAY (April 1)**

The Traditional
Failure:

**A manual system locks the 100 units for customer A.
Request B is Rejected (Revenue Lost).**

THE REASONING

STEP 1: Contextual Extraction *(1st April)*

AI reads Email B: "Needs 10 units today"

AI reads CSV Supplier: "Raw Material Lead Time = 5 days"

STEP 2: Multi-step Calculation

Current Stock: 100 units.

Action: Fulfill 10 units now → **Remaining:** 90 units.

Restock Plan: Trigger Purchase Order for 55 units Raw Material (5 units buffer).

Arrival Date: 1 April + 5 Days = **6 April.**

STEP 3: Temporal Buffer Check

Deadline (20 April) – Arrival (6 April) = 14 Days **SAFETY WINDOW**

Logic: The AI determines the 10-unit sale today has **Zero Impact** on the 20 April deadline.

THE OUTCOME

Decision: “Approve Request B + Initiate Procurement”

Autonomous Actions (No Manager Intervention):

- Generates **Purchase Order** to Supplier for raw materials.
- Updates **Production Schedule** to start on 7 April.

The Result:

Maximised Revenue

Both customers are satisfied
0% manual data entry

Deterministic Decision Rules

Drag to reorder priority — higher rules take precedence. Same inputs always produce the same outputs.

The agent evaluates rules top-to-bottom. The first matching condition triggers the action.

- 1 Budget Constraints HIGHEST
- 2 Urgent Customer Demandddd P2
- 3 Low Stock Replenishment P3
- 4 Supplier Delay P4
- 5 Production Blockage LOWEST

PO Budget Range

Set purchasing thresholds, spending caps, and reorder quantity policies.

Low Budget

Threshold **6500\$**

Maximum value for low-tier automatic PO approval.

1000\$ — 15000\$

Medium Budget

Threshold **29000\$**

Medium-tier POs — may require manager sign-off.

10000\$ — 60000\$

High Budget

Threshold **185000\$**

High-tier POs — always escalated for executive approval.

50000\$ — 200000\$

Dynamic Stock Allocation Rules

Control how the agent allocates, reallocates, and protects inventory across orders.

These rules govern real-time stock allocation decisions. Enabled rules are enforced deterministically during every allocation cycle.

- ☒ Prioritise orders by earliest deadline
Orders with the nearest due date receive stock allocation first, regardless of order size or customer.
- ☒ Allow reserved stock reallocation when safe
Permit the agent to move reserved stock between orders if it does not create a new stockout risk.
- ☒ Check manufacturing lead time before reallocating
Before reallocating stock, verify that a replacement manufacturing work order can be completed in time.
- ☒ Protect fulfilment KPI
Never allow a reallocation that would drop the overall fulfilment rate below the configured KPI target.
- ☒ Avoid delaying orders beyond committed deadlines
Block any stock movement that would push an existing order past its committed delivery date.
- ☐ Create replacement manufacturing work orders automatically
When stock is reallocated away from an order, automatically generate a new work order to replenish it.

Custom Allocation Rules

Provide any additional custom logic or instructions for the agent here.

e.g. Always source RAW-003 from secondary suppliers if primary lead time is over 10 days...

Supplier Preferences

Define how the agent ranks and selects suppliers for purchase orders.

- ☒ Preferred Supplier 35 %
- ☒ Lowest Cost 25 %
- ☒ Fastest Lead Time 25 %
- ☒ Reliability Score 15 %

Total Weight **100%**

KPI Preferences

Select which KPIs the agent should actively optimise during decision-making.

☒ Stockout Risk

☒ Fulfilment Rate

☒ Inventory Holding Cost

☒ Lead Time Reduction

☐ Budget Utilisation

4 of 5 KPIs active

Preferences UI

STRATEGIC GUARDRAILS

▶ Digital Twin

AI executes the business owner's specific strategy

▶ Dynamic Alignment

Reasoning shifts based on saved "personality"

▶ Set Once, Execute 1000x

SECURITY

▶ Source Verification

Cross-referencing CSV whitelists

▶ Sanity Checks

Blocking impossible orders (e.g., 1,000,000 units)

▶ Asset Protection

Defending against Social Engineering

BUSINESS IMPACT



MAXIMIZE REVENUE

Capture every
“Time-Sensitive” opportunity



RELIABILITY

Eliminated human
calculations errors in lead
times



STRATEGIC SCALABILITY

10x order volume on 1x
administrative cost



LABOUR

0% manual data entry –
Experts focus strategy